

Sandhya S

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Summary

Junior AI Engineer with experience in Artificial Intelligence, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Machine Learning. Skilled in LLM fine-tuning, prompt engineering, embeddings, vector databases, AI agents, tool calling, model optimization, and context engineering. Experienced in building scalable AI applications using Hugging Face Transformers, LangChain, LangGraph, and FastAPI.

Skills

Programming & Databases: Python, SQL, MySQL, DBMS

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Data Analysis

Machine Learning & Deep Learning: Scikit-learn, Machine Learning, Deep Learning, NLP, LLM Fine-Tuning, Hugging Face Transformers, PEFT, Unsloth, Ollama, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Model Evaluation, Embeddings, Vector Databases, Model Context Protocol (MCP), Pinecone, FAISS, Chroma

Frameworks & Deployment: PyTorch, TensorFlow, FastAPI, Docker, LangChain, LangGraph, Streamlit, Gradio

Cloud: AWS

Other Tools: Excel, Git, GitHub

Experience

Junior AI Engineer

Axomium Pvt Ltd

August 2024 – August 2026

Kerala, India

- Built and deployed domain-specific RAG applications using LangChain, LangGraph, Pinecone, and API-based LLMs, improving information retrieval accuracy by approximately 30% and reducing response latency by 40%.
- Fine-tuned multiple open-source LLMs using Hugging Face Transformers, LoRA, and PEFT on custom datasets containing 10,000+ samples, improving domain-specific response quality and reducing hallucinations.
- Built and deployed containerized FastAPI microservices and RAG pipelines using Docker, reducing deployment time by 50% and enabling scalable model serving.
- Applied model quantization and inference optimization techniques, reducing model memory usage by up to 60% and improving inference speed by 35% in resource-constrained environments.

Projects

OmniSight – Multimodal UI Self-Healing & RPA Agent

- Engineered a multimodal AI agent that detects UI issues from Playwright screenshots and HTML, generates fixes, modifies source code, and verifies repairs through automated retesting.
- Integrated Qwen VLM with FastAPI and a React dashboard for vision-based issue detection, automated fix generation, source modification, and verification.
- Executed browser automation and responsive UI testing across desktop, tablet, and mobile viewports using Playwright.

Music Recommender System

- Created a Natural Language Processing (NLP)-based recommendation system using TF-IDF and cosine similarity on over one million song lyrics.
- Integrated the Spotify API and launched a Streamlit web application, increasing recommendation accuracy by 30%.

VisioGuide

- Designed a lightweight smart cap that achieved 90% real-time object detection accuracy using TensorFlow-based Convolutional Neural Networks (CNNs).
- Developed voice-guided navigation using Pyttsx3 and optimized deployment on Raspberry Pi for improved portability and real-time performance.

Education

B.Tech in Computer Science and Engineering

APJ Abdul Kalam Technological University

October 2020 – June 2024

CGPA: 7.73 / 10